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Doctor of Engineering Student
Student Number: 20102931

July 31, 2026

Dr. Rosana J. Ferolin
ERDT Project Leader
Dean, School of Engineering

Through: Dr. Ramelito C. Agapay
ERDT Project Coordinator
Asst. Dean, School of Engineering

Subject: Request for a Second Scholarship Extension

Dear Dr. Ferolin,

I am writing to respectfully request an extension of my Engineering Research and Development for Technology (ERDT) scholarship, which is scheduled to conclude in July 2026.

I am a Doctor of Engineering student under the supervision of Dr. Lorafe F. Lozano. My dissertation was originally titled "Novel Machine Learning Methods for Process Optimization Problems in Biological Wastewater Treatment." In view of the progress and refinement of my research, I have narrowed its scope and revised the title to "Physics Enforcement on Machine Learning Surrogates of the Activated Sludge Process."

This revised title remains aligned with my approved dissertation topic while presenting a more focused scope. Specifically, the proposed novel machine learning method is physics enforcement on machine learning surrogates, together with an interpretable machine learning surrogate. The biological wastewater treatment process remains the activated sludge process, the work continues to be relevant to process optimization, and the model used is an improved version of the same bilinear model presented in my proposal manuscript.

To date, I have made substantial progress in my research. Most recently, my paper, "An Interpretable Statistical Surrogate of Activated Sludge Systems that Preserves Mass Conservation and Component Non-Negativity," was accepted for publication in the Q1 journal *Digital Chemical Engineering* (IF = 5.4). I have also submitted a second related paper to *Computers & Chemical Engineering* titled "Enforcement of Mass Conservation and Non-Negativity in Statistical Surrogates of Activated Sludge Systems."

Despite this progress, the complexity and innovative nature of my research have extended the timeline beyond both the original scholarship period and the initial extension period. The primary reasons for the delay are as follows:

- The process of publishing two articles in top-tier journals has required multiple rounds of review and revision, which extended beyond the initial one-year extension period.
- The specialized nature of the research required me to develop substantial expertise independently over time. As the sole author of the manuscripts submitted to top-tier chemical engineering journals, I was responsible for preparing the manuscripts and addressing reviewer comments independently.
- Since my scholarship stipends expired last year, I have had to engage in full-time work as a Senior Data Scientist to support myself while continuing my full-time academic research, which has made it challenging to balance both responsibilities.

Based on a detailed plan and timeline developed in consultation with my adviser, I respectfully request one final scholarship extension until July 2027. This extension will provide the necessary time to complete the final stages of my research publication, finalize my dissertation manuscript, and prepare for and complete my thesis defense.

I fully understand and acknowledge that no new financial assistance—including stipends, tuition fees, or other allowances—will be disbursed during the extension period. I also understand that I may only avail myself of the remaining grants from my original allocation.

For your review, the following required documents are attached to this letter:

1. ISMIS Grade with CGWA;
2. Approved Thesis Proposal Approval Sheet;
3. Detailed list of activities and Gantt chart, certified by my adviser;
4. Certification from my adviser regarding the expected date of my thesis defense; and
5. Certification from my adviser regarding the expected date of graduation.

Thank you very much for your time and kind consideration of my request.

I remain committed to completing my degree to the highest standard and contributing meaningfully to the objectives of the ERDT program.

Sincerely,


Egberto F. Selerio Jr.

07/31/2026